Experiment 7

Aim: Add multimedia functionality to your mobile application, such as capturing photos/videos, playing audio files, or integrating with social media sharing.

Theory:

Multimedia functionality in mobile applications allows users to interact with images, audio, videos, and sharing features, making applications more interactive and user-friendly. Flutter provides various packages to integrate multimedia features such as camera access, gallery selection, audio playback, and social media sharing.

In this experiment, a multimedia application is developed using Flutter. The application allows users to capture images using the camera, select images from the gallery, play audio files, and share content using the device's sharing options. The `image_picker` package is used for camera and gallery access, `audioplayers` is used for audio playback, and `share_plus` enables content sharing.

The application also includes audio controls such as play, pause, seek, and duration tracking to improve user experience. This experiment demonstrates how multimedia integration enhances the functionality and interactivity of mobile applications.

Source Code:

```
import 'dart:io';
import 'dart:typed_data';
import 'package:flutter/foundation.dart';
import 'package:flutter/material.dart';
import 'package:image_picker/image_picker.dart';
import 'package:audioplayers/audioplayers.dart';
import 'package:share_plus/share_plus.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        brightness: Brightness.light,
        colorScheme: ColorScheme.fromSeed(seedColor:
Colors.blue),
      ),
      home: MultimediaPage(),
    );
  }
}

class MultimediaPage extends StatefulWidget {
  @override
  _MultimediaPageState createState() =>
_MultimediaPageState();
}

class _MultimediaPageState extends State<MultimediaPage> {
  File? _image;
  Uint8List? _webImage;
```

```
  final ImagePicker _picker = ImagePicker();
  final AudioPlayer _player = AudioPlayer();

  bool isPlaying = false;
  Duration duration = Duration.zero;
  Duration position = Duration.zero;

  @override
  void initState() {
    super.initState();

    // Audio duration
    _player.onDurationChanged.listen((d) {
      setState(() => duration = d);
    });

    // Audio position
    _player.onPositionChanged.listen((p) {
      setState(() => position = p);
    });

    // Audio complete
    _player.onPlayerComplete.listen((event) {
      setState(() {
        isPlaying = false;
        position = Duration.zero;
      });
    });
  }

  // Camera
  Future<void> captureImage() async {
    if (kIsWeb) {
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text("Camera not supported on
FlutLab Web")),
      );
      return;
```

```
    }

    final picked = await _picker.pickImage(source:
ImageSource.camera);

    if (picked != null) {
      setState(() {
        _image = File(picked.path);
      });
    }
  }

  // Gallery
  Future<void> pickImage() async {
    final picked = await _picker.pickImage(source:
ImageSource.gallery);

    if (picked != null) {
      if (kIsWeb) {
        final bytes = await picked.readAsBytes();
        setState(() {
          _webImage = bytes;
        });
      } else {
        setState(() {
          _image = File(picked.path);
        });
      }
    }
  }

  // Audio Play/Pause
  Future<void> toggleAudio() async {
    if (!isPlaying) {
      await _player.play(
        UrlSource(
```

```
'https://www.soundhelix.com/examples/mp3/SoundHelix-Song-
1.mp3'),
      );
      setState(() => isPlaying = true);
    } else {
      await _player.pause();
      setState(() => isPlaying = false);
    }
  }

  // Format Time
  String formatTime(Duration d) {
    String twoDigits(int n) => n.toString().padLeft(2, '0');
    final minutes = twoDigits(d.inMinutes);
    final seconds = twoDigits(d.inSeconds % 60);
    return "$minutes:$seconds";
  }

  // Share
  void shareContent() {
    Share.share("Check out my Flutter Multimedia App 🚀");
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Multimedia App"),
        centerTitle: true,
        elevation: 0,
      ),
      body: Container(
        decoration: BoxDecoration(
          gradient: LinearGradient(
            colors: [Colors.blue.shade200, Colors.white],
            begin: Alignment.topCenter,
            end: Alignment.bottomCenter,
```

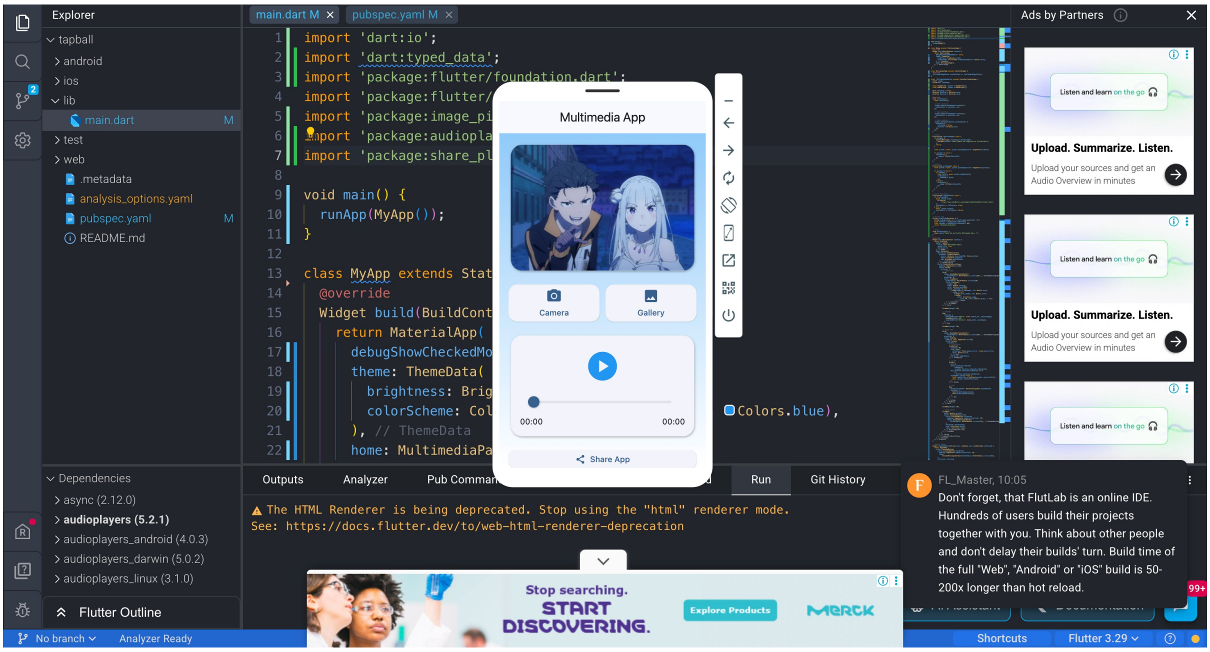
```
          ),
        ),
        child: SingleChildScrollView(
          padding: EdgeInsets.all(16),
          child: Column(
            children: [
              // IMAGE CARD
              Card(
                shape: RoundedRectangleBorder(
                    borderRadius: BorderRadius.circular(20)),
                elevation: 5,
                child: ClipRRect(
                  borderRadius: BorderRadius.circular(20),
                  child: Container(
                    height: 220,
                    width: double.infinity,
                    color: Colors.grey[200],
                    child: _webImage != null
                        ? Image.memory(_webImage!, fit:
BoxFit.cover)
                        : _image != null
                            ? Image.file(_image!, fit:
BoxFit.cover)
                            : Center(
                                child: Icon(Icons.image,
                                    size: 60, color:
Colors.grey),
                              ),
                  ),
                ),
              ),

              SizedBox(height: 20),

              // BUTTONS
              Row(
                children: [
```

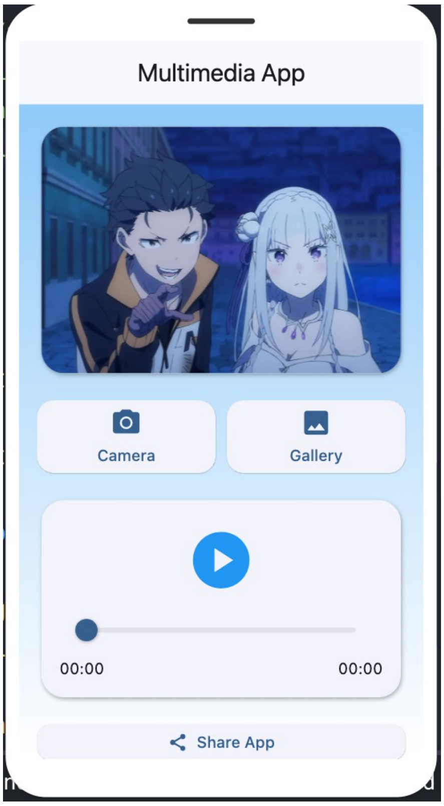
```
                  expandedButton("Camera", Icons.camera_alt,
captureImage),
                  SizedBox(width: 10),
                  expandedButton("Gallery", Icons.photo,
pickImage),
                ],
              ),

              SizedBox(height: 20),

              // AUDIO CARD
              Card(
                shape: RoundedRectangleBorder(
                    borderRadius: BorderRadius.circular(20)),
                elevation: 5,
                child: Padding(
                  padding: const EdgeInsets.all(16),
                  child: Column(
                    children: [
                      // Play Button
                      IconButton(
                        iconSize: 60,
                        icon: Icon(
                          isPlaying ? Icons.pause_circle :
Icons.play_circle,
                          color: Colors.blue,
                        ),
                        onPressed: toggleAudio,
                      ),

                      // Slider
                      Slider(
                        min: 0,
                        value: position.inSeconds
                            .toDouble()
                            .clamp(0,
duration.inSeconds.toDouble()),
```

```
                        max: duration.inSeconds.toDouble() ==
0
                            ? 1
                            : duration.inSeconds.toDouble(),
                        onChanged: (value) async {
                          final newPosition =
Duration(seconds: value.toInt());
                          await _player.seek(newPosition);
                        },
                      ),

                      // Time
                      Row(
                        mainAxisAlignment:
MainAxisAlignment.spaceBetween,
                        children: [
                          Text(formatTime(position)),
                          Text(formatTime(duration)),
                        ],
                      ),
                    ],
                  ),
                ),
              ),

              SizedBox(height: 20),

              // SHARE BUTTON
              SizedBox(
                width: double.infinity,
                child: ElevatedButton.icon(
                  style: ElevatedButton.styleFrom(
                    padding: EdgeInsets.symmetric(vertical:
14),
                    shape: RoundedRectangleBorder(
                        borderRadius:
BorderRadius.circular(12)),
                  ),
```

```
                  onPressed: shareContent,
                  icon: Icon(Icons.share),
                  label: Text("Share App"),
                ),
              ),
            ],
          ),
        ),
      ),
    );
  }

  // Button Widget
  Widget expandedButton(String text, IconData icon,
VoidCallback onPressed) {
    return Expanded(
      child: ElevatedButton(
        style: ElevatedButton.styleFrom(
          padding: EdgeInsets.symmetric(vertical: 14),
          shape:
              RoundedRectangleBorder(borderRadius:
BorderRadius.circular(15)),
        ),
        onPressed: onPressed,
        child: Column(
          children: [
            Icon(icon, size: 28),
            SizedBox(height: 5),
            Text(text),
          ],
        ),
      ),
    );
  }
}
```

Output:





Learning Outcomes: